

Flight Arrival Delay Prediction: A Multi-Class Severity Approach

Gracie Nguyen, Leah Ho, Nam Nguyen

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ABSTRACT

Using BTS on-time data, this project predicts how severely flights from Harrisburg International (MDT) will be delayed. KMeans clustering sorted delays into four classes — on-time, minor, moderate, and severe — based on actual delay distributions. Three models (Logistic Regression, Random Forest, XGBoost) were trained on features like carrier, destination, departure hour, day of week, distance, and prior-leg delay status.

MOTIVATION

Which model yields the best result when it comes to predicting arrival delays for flights departing from MDT?

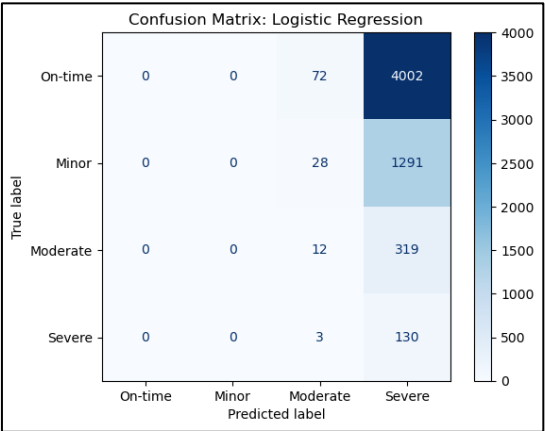
DATA & METHODS

Dataset: Bureau of Transportation Statistics (BTS)
Scope: Approx. 29,281 departures from MDT
Target: Four levels of delay severity (On-Time, Minor, Moderate, Severe)
Method: Stratified train/test split

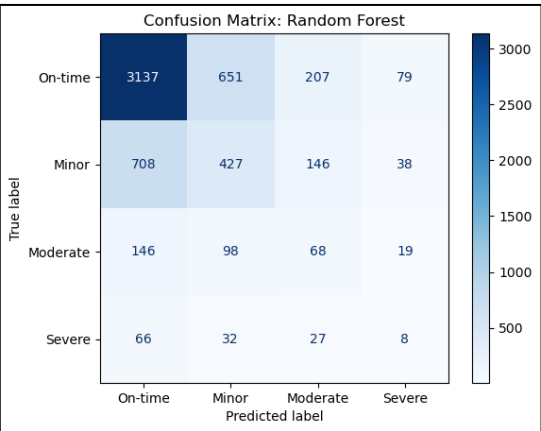
FEATURE ENGINEERING

```
<class 'pandas.core.frame.DataFrame'>
Index: 29281 entries, 1087201 to 809569
Data columns (total 17 columns):
 #   Column              Non-Null Count  Dtype  
---  -
 0   YEAR                29281 non-null  int64  
 1   MONTH              29281 non-null  int64  
 2   DAY_OF_MONTH       29281 non-null  int64  
 3   DAY_OF_WEEK        29281 non-null  int64  
 4   CRS_DEP_TIME       29281 non-null  int64  
 5   OP_UNIQUE_CARRIER 29281 non-null  object  
 6   DEST               29281 non-null  object  
 7   DISTANCE           29281 non-null  float64 
 8   CRS_ELAPSED_TIME   29281 non-null  float64 
 9   TAIL_NUM           29281 non-null  object  
10   FL_DATE            29281 non-null  datetime64[ns]
11   ARR_DEL15          29281 non-null  int64  
12   ARR_DELAY_NEW      29281 non-null  int64  
13   DEP_HOUR           29281 non-null  int64  
14   PREV_FLIGHT_DELAYED 29281 non-null  int64  
15   ARR_DELAY_CAPPED   29281 non-null  int64  
16   DELAY_CLUSTER      29281 non-null  int64  
dtypes: datetime64[ns](1), float64(2), int64(11), object(3)
memory usage: 5.0+ MB
```

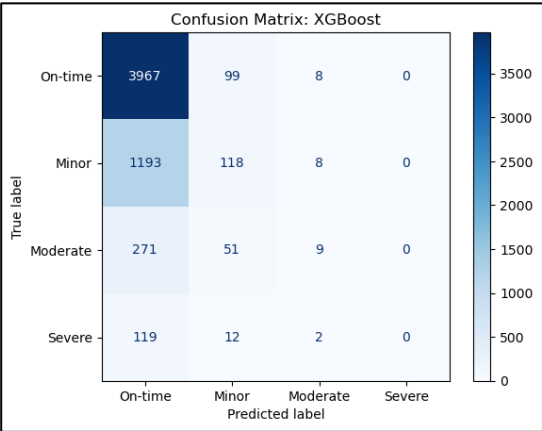
LOGISTIC REGRESSION (LR)



RANDOM FOREST (RF)



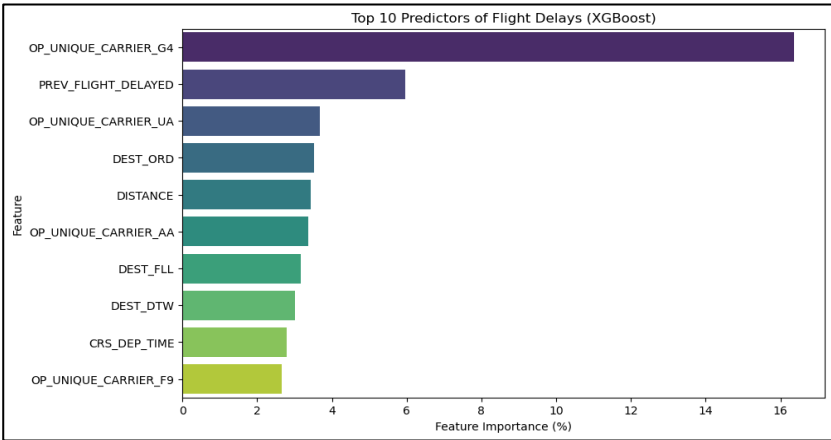
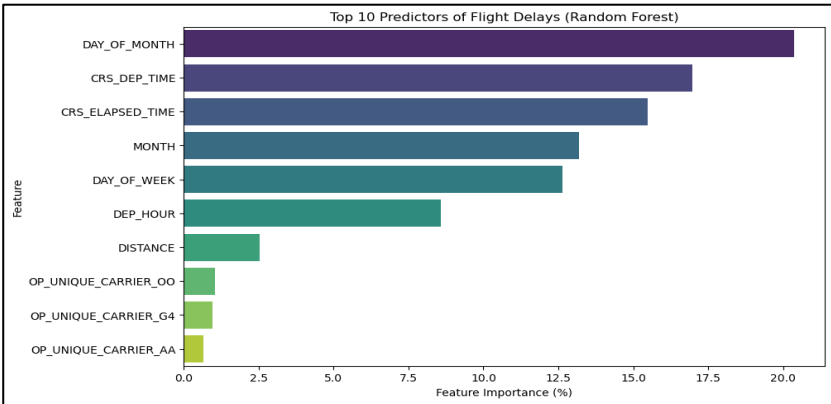
XGBOOST (XGB)



LIMITATIONS

- Accurate Severe delay prediction remained difficult
- Minority class recall remained relatively low
- Weather variables were not integrated
- Airport congestion variables were unavailable
- Advanced resampling techniques such as SMOTE were not fully explored
- Historical delay data by airline was not exploited

FEATURES IMPORTANCE



Results	LR	RF	XGB
Accuracy	0.0242	0.6215	0.6990
Weighted F1	0.0040	0.6240	0.6095
Weighted ROC-AUC	0.5872	0.6626	0.6869
Weighted Average Precision	0.5918	0.6447	0.6648

CONCLUSION

Ensemble methods substantially outperformed Logistic Regression, which nearly collapsed across all metrics. XGBoost led in accuracy (0.699), weighted ROC-AUC (0.6869), and weighted average precision (0.6648), making it the stronger model for overall discrimination. Random Forest, however, achieved the highest weighted F1-score (0.6240), suggesting better balance across classes, making it the preferred choice when correctly identifying serious delay categories is the operational priority.

REFERENCE

Snell, Kenney, et al. "Predicting Flight Delays Using Machine Learning." (2026).

Gui, Guan, et al. "Flight delay prediction based on aviation big data and machine learning." *IEEE Transactions on Vehicular Technology* 69.1 (2019): 140-150.